

Annihilator of a Sum of Subspaces

Linear Algebra

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Problem Statement

Problem

Let W_1 and W_2 be subspaces of a finite-dimensional vector space V . Describe $A(W_1 + W_2)$ in terms of $A(W_1)$ and $A(W_2)$, where $A(W)$ denotes the annihilator of W .

Target Result

$$A(W_1 + W_2) = A(W_1) \cap A(W_2)$$

Proof: Part 1

Forward Inclusion: $A(W_1 + W_2) \subseteq A(W_1) \cap A(W_2)$

Since W_1 and W_2 are subspaces of $W_1 + W_2$, we have:

$$W_1 \subseteq W_1 + W_2 \quad \text{and} \quad W_2 \subseteq W_1 + W_2$$

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Reversing inclusions via annihilators yields:

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Combining both statements:

$$A(W_1 + W_2) \subseteq A(W_1) \cap A(W_2) \tag{1}$$

Proof: Part 2

Reverse Inclusion: $A(W_1) \cap A(W_2) \subseteq A(W_1 + W_2)$

Let $f \in A(W_1) \cap A(W_2)$:

$$\implies f \in A(W_1) \quad \text{and} \quad f \in A(W_2)$$

$$\implies f(w_1) = 0 \quad \text{and} \quad f(w_2) = 0 \quad \forall w_1 \in W_1, w_2 \in W_2$$

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Now, let $w \in W_1 + W_2$. Then $w = w_1 + w_2$ for some $w_1 \in W_1$ and $w_2 \in W_2$:

$$f(w) = f(w_1 + w_2) = f(w_1) + f(w_2) = 0 + 0 = 0$$

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Since $f(w) = 0$ for all $w \in W_1 + W_2$:

$$f \in A(W_1 + W_2) \implies A(W_1) \cap A(W_2) \subseteq A(W_1 + W_2) \quad (2)$$

Final Result

From inclusions (1) and (2):

$$A(W_1 + W_2) \subseteq A(W_1) \cap A(W_2)$$

$$A(W_1) \cap A(W_2) \subseteq A(W_1 + W_2)$$

Therefore:

$$A(W_1 + W_2) = A(W_1) \cap A(W_2)$$